Let us list out the half angle identities in the following table:

Double angle identity	Half-angle identity
$\sin 2A = 2\sin A\cos A$	$\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$
$\cos 2A = \cos^2 A - \sin^2 A$	$\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$
$\cos 2A = 2\cos^2 A - 1$	$\cos\theta = 2\cos^2\frac{\theta}{2} - 1$
$\cos 2A = 1 - 2\sin^2 A$	$\cos\theta = 1 - 2\sin^2\frac{\theta}{2}$
$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$	$\tan\theta = \frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$
$\sin 2A = \frac{2\tan A}{1+\tan^2 A}$	$\sin\theta = \frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$
$\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}$	$\cos\theta = \frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

(i) The half angle identities are often used to replace a squared trigonometric function by a non squared trigonometric function.

(ii) Half angle identities allow us to find the value of the sine and cosine of half the angle if we know the value of the cosine of the original angle.

Example 3.22 Find the value of $\sin\left(22\frac{1}{2}^\circ\right)$

Solution:

We know that $\cos\theta = 1 - 2\sin^2\frac{\theta}{2} \Rightarrow \sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}$. Take $\theta = 45^\circ$

we get $\sin\frac{45^\circ}{2} = \pm\sqrt{\frac{1-\cos 45^\circ}{2}}$, (taking positive sign only, since $22\frac{1}{2}^\circ$ lies in the first quadrant)

Thus, $\sin 22\frac{1}{2}^\circ = \sqrt{\frac{1-\frac{1}{\sqrt{2}}}{2}} = \frac{\sqrt{2-\sqrt{2}}}{2}$.

Example 3.23 Find the value of $\sin 2\theta$, when $\sin\theta = \frac{12}{13}$, θ lies in the first quadrant.

Solution:

Using a right triangle, we can easily find that $\cos\theta = \frac{5}{13}$

$\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{12}{13}\right)\left(\frac{5}{13}\right) = \frac{120}{169}$.